

Scientific Miracles of the Qur'ān

Qur'ānic Integrity and Scientific Advancement

A miracle is a divine act which defies and transcends universal laws which God grants to the Prophets in order to convince people of the truth of their message. The Qur'ān uses the word 'Āyat' [literally meaning 'sign'] to describe this phenomenon. These Divine miracles are designed to provide people with the capacity to perceive them as signs of God's power, knowledge and will over all things in the universe. For a miracle to have full impact it should relate to an area of knowledge in which its intended audience are well versed; in this way can a miracle fulfill its purpose.

Historically, every Prophet was given miracles and these were witnessed by their respective contemporaries. The Prophet Muhammad is unique in the chain of Prophethood, for unlike all previous Prophets, He was sent as God's final Prophet and Messenger for all mankind. Reason demands, therefore, that he should have a universal miracle that is not bound by time or geographical considerations. Every individual at every stage of human history, no matter where he lived in the world, is hence justified in saying, "If Muhammad is a Prophet for me today. I would like to behold a miracle today."

Unlike the tangible miracles [an example being the splitting of the sea] of the previous Prophets, the main miracle of the Prophet Muhammad, although he was given other miracles, was an intellectual miracle. The effectiveness of the tangible miracles would have their greatest impact on eye witnesses and their full impact would end with the death of these witnesses. If we ask a Jew or Christian to show us the miracles of the Prophet Moses or Jesus – they both would submit that it is not within human power to demonstrate any of those miracles now. Moses cannot be asked to split the sea again and the Prophet Jesus cannot be called to raise people from the dead. For us today, these miracles are nothing more than historical reports. But if a Muslim is asked about the greatest miracle of the Prophet Muhammad, he can readily show His Book, the Qur'ān – for the Qur'ān is a miracle which remains in our hands. It is a Book for all people to examine its contents.

The miraculous nature of the Qur'ān lies in the knowledge it contains and this is everlasting, renewable and a living proof. The Qur'ān was given to mankind more than fourteen centuries ago. Recent advancements in various sciences have revealed innumerable facts. None of these discoveries contradict a single verse of the Qur'ān. In fact, each time the world makes a breakthrough in unravelling the universe, we increase in certainty that the One who created this universe is the One who revealed the Qur'ān. The Qur'ān is filled with important facts which were revealed at a point in human history when no man could possibly have known them. The Qur'ān speaks about the origins of the universe, workings of celestial bodies, stages of embryonic human development, behavioural patterns of species, historical and archaeological truths, etc.

Non-believers will naturally seek an explanation as to the source of knowledge of the Qur'ān. Atheists would argue that the Qur'ān was the product of intellectual ramblings of the Prophet Muhammad and the Jews and Christians would say that it was plagiarised from the Old and New Testament. The fact is that such statements do not belong to the period when the Qur'ān was revealed and no such statements exist in the Old and New Testament. It is obvious that no one man could possess all the knowledge that is contained in the Qur'ān. One could conceivably be lucky with a few guesses, but the variety and quality of accurate knowledge in the Qur'ān shows, without

doubt, these are not the words of a mere mortal. No human being has ever written a comprehensive book with such absolute perfection and certainty of knowledge. Every book represents the knowledge available to a writer at a certain time and place, and human knowledge keeps advancing. As a result, every book written by a human being becomes obsolete after some time because the information is found to be either false or incomplete, raising the need for books to be revised or re-written. The writings of every physicist, scientist, biologist or philosopher have proved this to be the case.

The Qur'ān contains over 6,000 Ayat [verses] with over 70,000 words, yet it is a fact that not a single word, phrase, sentence or topic needs to be revised. It is also the case that a writer usually has knowledge of a specific subject or a few related subjects. For example, one can write about history or economics or philosophy or physics etc. But it is not possible for a human being to write simultaneously on a great variety of different subjects with such in-depth and accurate information. Even a casual reader finds that the Qur'ān discusses an array of subjects with the accuracy and precision to make them appear to be the observations of an eye witness expert.

Finally, it must be clearly understood that the Qur'ān is not a book of science, philosophy, geology or history - it is a revelation from God for the purpose of guiding Mankind. Nonetheless, if any expert from the various fields of knowledge analysed what the Qur'ān states; they would conclude that the Qur'ān, beyond doubt, is not authored by a man in the 7th century. If what the Qur'ān imparted was unfounded, science would, in time, have proved its falsehood and the whole religion would have been destroyed. On the contrary, every scientific theory, from the historic past, that contradicted the Qur'ān has been proven to be unfounded and lacking empirical evidence. For each successive generation, the Qur'ān offers new and relevant meanings, proportionate to the development of knowledge and growth of that generation's intellectual aptitude. In other words, they are applicable to the constantly developing perceptions of the human race, expanding in meaning as man's exploration of the universe and his search for knowledge about his life and existence increases. From this we see that the Qur'ān was not intended solely for one people or nation, but came for the benefit and guidance of all; a complete and comprehensive religion, providing knowledge for all the generations. If these teachings were limited to one century or generation, the everlasting objective of the Qur'ān would have long since diminished. Yet the Qur'ān is constantly regenerating new meaning, providing a continual source of guidance.

The Qur'ān on the 'Big Bang Theory'

Historic Preamble

The enormously vast universe has been the object of curiosity since time immemorial. Greek philosophers, including Aristotle, believed that the Universe had always existed and would continue to do so eternally. This was also the mainstream view in scientific circles at the beginning of the 20th century, aptly known as the 'steady state theory.' An eternal state of the universe meant that there was no inherent need for a Creator – for what does not have a beginning does not necessitate a need for a cause. However, advancements in science would shatter this view and fundamentally prove that the Universe had a beginning.

In 1922, physicist Alexander Friedmann, produced computations showing that the structure of the universe was not static and that even a tiny impulse might be sufficient to cause the whole structure

to expand or contract according to Einstein's 'Theory of General Relativity.' George Lemaitre was the first to recognise the implications of what Friedman concluded. Based on these computations, Lemaitre declared that the universe must have had a beginning and that it was expanding as a result of something that triggered it. He also stated that the rate of radiation could be used as a measure of the aftermath of that 'something.'

The theoretical musings of these two scientists did not attract much attention and probably would have gone ignored except for new observational evidence that rocked the scientific world in 1929. That year, American astronomer Edwin Hubble, made one of the most important discoveries in the history of astronomy. He discovered that galaxies were moving away from us at speeds directly relative to their distance from us and from each other.

A universe where everything constantly moves away from everything else implied a constantly expanding universe. Stephen Hawking writes, "The expansion of the universe was one of the most important intellectual discoveries of the 20th century, or of any century."

Since the universe is constantly expanding, were we to rewind a film [of its history], then necessarily we would find the entire universe was in a joint state, referred to by some as the 'Primordial Atom.' Many scientists and philosophers resisted the idea of a beginning to the universe because of the many questions that it raised – primarily what or who caused it. However, with Penzias and Wilson's discovery of microwave radiation emanating from all directions, possessing the same physical characteristics - namely petrified light which came from a huge explosion during the first seconds after the birth of the universe – left little doubt about the fact that the universe had a beginning.

For fourteen hundred years, since the revelation of the Qur'ān, sceptics had trouble understanding the verse, **'...the heavens and the earth were a joined entity and We separated them...'** [21:30]. However, with the assistance of scientific advancements, we can now understand these verses in a new light which help us piece together the cosmological puzzle. The miraculous nature of the Qur'ān lies in the knowledge it contains. Its verification of scientific facts shows that its message is as applicable to the scientist in his laboratory today as it was to the Bedouin in the desert.

Linguistic Analysis

"Have those who disbelieved not considered that the heavens and the earth were a [ratq] joined entity, and We [fataqa] separated them....Then will they not believe?"
Qur'ān 21:30

The word '*ratq*' translated as 'sewn to' means 'mixed in each, blended' in Arabic. It is used to refer to two different substances that make up a whole. The phrase '*fataqa*' is 'unstitched' and implies that something comes into being by tearing apart or destroying the structure of things that are sewn to one another.

In the verse, heaven and earth are at first subject to the status of '*ratq*.' They are separated [*fataqa*] with one coming out of the other. Intriguingly, when we think about the first moments of the 'Big Bang' we see that the entire matter of the universe collected at one single point. In other words, everything including 'the heavens and earth' which were not created yet were in an interwoven and inseparable condition. Then, this point exploded violently, causing its matter to disunite.

The Qur'ān on the Origin of the Universe

The science of modern cosmology, observational and theoretical, clearly indicates that, at one point in time, the whole universe was nothing but a cloud of 'smoke' [i.e. an opaque highly dense and hot gaseous composition].¹⁹⁰⁷ This is one of the undisputed principles of standard modern cosmology. Scientists now can observe new stars forming out of the remnants of that 'smoke' [see Figures 1 and 2].



Figure 1: A new star forming out of a cloud of gas and dust (nebula), which is one of the remnants of the 'smoke' that was the origin of the whole universe. (*The Space Atlas, Heather and Henbest, p. 50.*)



Figure 2: The Lagoon nebula is a cloud of gas and dust, about 60 light years in diameter. It is excited by the ultraviolet radiation of the hot stars that have recently formed within its bulk. (*Horizons, Exploring the Universe, Seeds, plate 9, from Association of Universities for Research in Astronomy, Inc.*)

The illuminating stars we see at night were, just as was the whole universe, in that 'smoke' material. God has said in the Qur'ān:

“Then He directed Himself to the heaven while it was smoke...”

Qur'ān 41:11

Because the earth and the heavens above (the sun, the moon, stars, planets, galaxies, etc.) have been formed from this same 'smoke,' we conclude that the earth and the heavens were one connected entity.

We know that our world, the sun and the stars did not come about immediately after the primeval explosion. For the universe was in a gaseous state before the formation of the stars. This gaseous state was initially made of hydrogen and helium. Condensation and compression shaped the planets, the earth, the sun and the stars that were but products of the gaseous state. The discovery of these phenomena has been rendered possible thanks to successive findings as a result of observations and theoretical developments.

The knowledge of all contemporary communities during the time of the Prophet would not suffice for the assertion that the universe had once been in a gaseous state. The Prophet himself did not

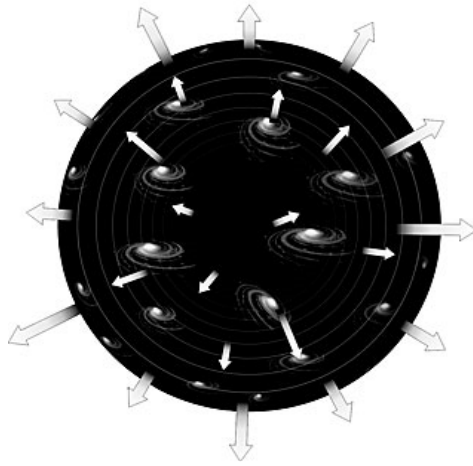
¹⁹⁰⁷ Weinberg, *The First Three Minutes, a Modern View of the Origin of the Universe*, pp. 94-105.

claim to be the author of the statements in the Qur'ān as it often reminded, declaring that he is simply a messenger of God.

“That is from the news of the unseen which We reveal to you, [O Muhammad]. You knew it not, neither you nor your people, before this. So be patient; indeed, the [best] outcome is for the righteous.”

Qur'ān 11:49

The Qur'ān on the Expanding Universe



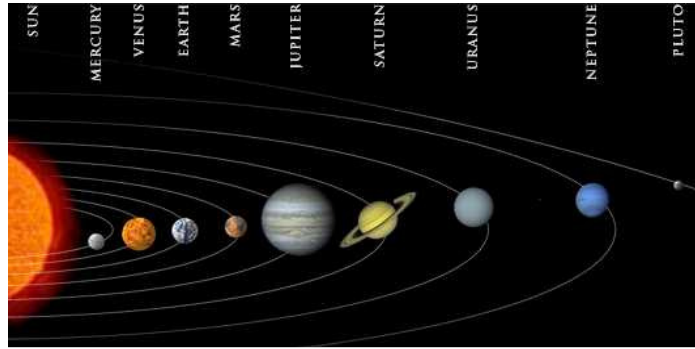
“And the heaven We constructed with strength, and indeed, We are [its] expander.”

Qur'ān 51:47

It was only after the development of the radio telescope in 1937, that the expansion of the universe was observed and established. While observing the sky with a telescope, Edwin Hubble, the American astronomer, discovered that the stars and galaxies were constantly moving away from each other. This discovery is regarded as one of the greatest in the history of astronomy. During these observations, Hubble established that the stars emit a light that turns redder according to their distance.

That is because according to the known laws of physics, light heading towards a point of observation turns violet, and light moving away from that point assumes a more reddish hue. During his observations, Hubble noted a tendency towards the colour red in the light emitted by stars. In short, the stars were moving further and further away, all the time. The stars and galaxies were not only moving away from us, but also from each other. A universe where everything constantly moves away from everything else implied a constantly expanding universe. The debate now is not whether the universe is expanding but rather at what rate. In 2011, the Nobel Prize in Physics was awarded to three scientists for the ‘discovery of the accelerating expansion of the universe through observations of distant supernovae.’

The Qur'ān on the Orbital Movement of the Sun and the Moon



“And it is He who created the night and the day and the sun and the moon; all [heavenly bodies] in an [*falak*] orbit are [*yasbahoona*] swimming.”

Qur'ān 21:33

“It is not allowable [i.e., possible] for the sun to reach the moon, nor does the night overtake the day, but each, in an [*falak*] orbit, is [*yasbahoona*] swimming.”

Qur'ān 36:40

The Arabic words used in these verses are *falak* and *yasbahoona* which can be translated as ‘sphere or orbit’ and ‘swimming.’ This concept of the movement of the sun and the moon and the other planets is in perfect harmony with recent discovery. It is inconceivable that an Arab, living centuries ago in the most primitive part of the world, could have rightly used such a specific term to describe the movements of planets without divine guidance. It should be noted that the discovery of the orbital movement of all celestial bodies was due to the invention of telescopes.

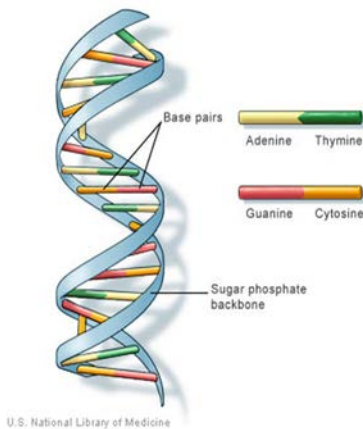
The Qur'ān on Duality in Creation

“Exalted is He who created all pairs - from what the earth grows and from themselves and from that which they do not know.”

Qur'ān 36:36

This Qur'ānic verse outlines the fact that all creatures, whether living beings or solid matter, are created in pairs. It refers to everything that was created. Amazingly, the outstanding truth and generality of this and similar verses came to be gradually realised, and more so recently, during 14 centuries since the Qur'ān was first revealed in a primitive world.

Millions of animal species discovered, classified and investigated only during the last two centuries, were found to be invariably in ‘pairs,’ male and female. Electron microscopy has clarified that all living creatures, however minute, are in pairs. The smallest microbes, viruses, and bacteria have their counterpart antibodies.



Take for example DNA – it is made up of thousands of different genes, and genes are made up of base pairs. These ‘base pairs’ are made of two paired up nucleotides. In order to form a base pair, we need to pair up specific nucleotides. Each type of nucleotide has a specific shape, so only certain combinations fit.

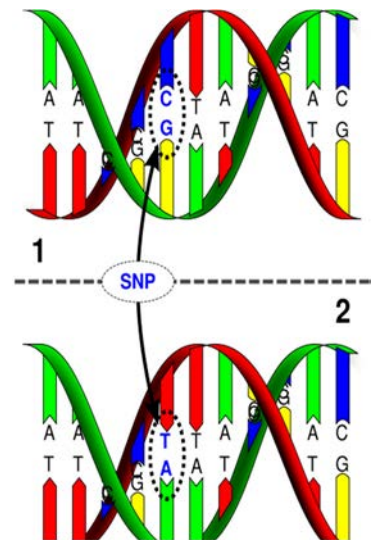
The sequence, composition, and orientation of these ‘pairs’ of nucleotides control the genetic information carried by the DNA. A chromosome consists of different types of protein bound tightly with a single DNA molecule chain.

The DNA is a large long (up to 1 meter long) amino acid chain. It consists of a ‘pair’ of spiral strands, connected with steps. Each step consists of a ‘pair’ of chemical components, so-called nucleotides.

There are 4 nucleotides. Adenine, Thymine, Guanine and Cytosine represented respectively by the letters A,T,G and C. Due to their shapes only A and T or G and C fit into one another.

Base Pairs [A-T, G-C] (billions of these **matching pairs**) --->
 Genes (thousands of these) --> DNA --> Chromosomes -->
 Nucleotides --> Nucleus (the ‘brain’ of the cell).¹⁹⁰⁸

All life systems including plant, animal and human consist of different types of cells. A cell consists of a nucleus surrounded with cytoplasm which is usually enclosed, within a cell wall. The cell nucleus, carries the chromosomes that control all the cell functions. All cells of a particular organism have exactly the same number of chromosomes; the number varies widely between different species. Proteins are formed from various combinations of amino acids. Specifically, 20 types of amino acid are used in different combinations to form more than a million types of protein, present in a human being. Every type of amino acid can exist in either of a pair of structures (right-handed isomer or left-handed isomer), with opposite polarised light rotation direction. The same applies to the proteins formed thereof.



The wide variety of creatures including living species, solid matter, liquids and gases are marvellous combinations of the same list of building blocks: atoms. These basic units, were long known to

¹⁹⁰⁸ International Human Genome Sequencing Consortium (2004). “Finishing the euchromatic sequence of the human genome.” Nature 431 (7011): 931–45

consist of a 'pair' of a positively charged nucleus surrounded by negative electrons. The nucleus consists of protons that carry the positive charge, together with neutrons. Even the neutral neutrons have their counterpart, the anti-neutrons. Later advances in nuclear physics has demonstrated that each of these particles is, in effect, a complex structure of much smaller nuclear particles. Over 200 of such elementary particles are now known.

At the atomic level, atoms can, literally, ionise i.e. either lose or gain electrons to form positive cations or negative anions. 'Pairs' of cations/anions combine to produce the wide variety of chemical (inorganic) compounds. This is one of the conclusions made by British physicist Paul Dirac, winner for Nobel Prize for Physics in 1933. His finding, known as 'parity,' revealed the duality known as matter and anti-matter.

Another example of duality in creation is plants. Botanists only discovered that there is a gender distinction in plants some 100 years ago. Yet, the fact that plants are created in pairs was revealed in the verses of the Qur'ān 1,400 years ago. It was only after the discovery of microscopes that human beings knew that plants have male organs (stamens) and female organs (ovaries) and that the wind, together with other factors, carries the pollen from one type to the opposite one so that reproduction can take place.

Every animal species of the wide animal kingdom reproduce sexually. Sexual reproduction results from the combination of a female ovum and a male sperm. The formation of this zygote 'pair' is the starting point in the reproduction cycle. The sperms, in turn are of 'two' kinds, the first carries the hereditary male characteristics, while the other carries the female ones.

Flowering plants, of which more than 250,000 have been discovered so far, also reproduce sexually. They have both female (ovaries containing eggs) and male (stamens carrying pollens); either combined in the same flower or in different flowers. In the latter case, fertilization occurs when pollens are transferred by wind or insects to an adjacent flower.

Non-flowering plants, on the other hand, amounting to 150,000 species, reproduce in a double-stage cycle of sexual and asexual reproduction. Yet, the asexual reproduction stage is essentially a process of breaking up the DNA 'pair' of strands into two; followed by each of which re-forming its complementary strand. Thus, a new 'pair' of identical DNA molecules results in the cell, just before it divides into a 'pair' of identical cells. The same applies to the asexual reproduction of bacteria.

Each bacterium consists of a single cell, the smallest biological unit able to function independently. A single bacterium reproduces the same way explained above, i.e. by splitting into a pair of identical cells. As we have seen, cell division occurs through the process of DNA replication, in which the two strands of the DNA molecules are separated; and each strand resynthesizes a complementary strand to itself. So, 'asexual' reproduction of bacteria involves the DNA 'pair' of strands splitting and reformation into a new 'pair' of cells.

The Qur'ān on the Origin of Life in Water

“Have those who disbelieved not considered that the heavens and the earth were a joined entity, and We separated them and made from water every living thing? Then will they not believe?”

Qur'ān 21:30

The origin of life is now such a basic scientific fact that it is accepted without hesitation. This could lessen one's appreciation for these verses. Yet it must be borne in mind that the Arabian peninsula is a desert land without a single lake or river, these verses describe something unimaginable to those at the time of the Prophet Muhammad.

A point to note - animals in dry regions have been created with mechanisms to protect their metabolisms from water loss and to ensure maximum benefit from water use. If water loss takes place in the body for any reason, and if that loss is not made good, death will result in a few days.

The Qur'ān on Seas and Rivers

Modern Science has discovered that in the places where two different seas meet, there is a barrier between them. This barrier divides the two seas so that each sea has its own temperature, salinity, and density.¹⁹⁰⁹ For example, Mediterranean sea water is warm, saline, and less dense, compared to Atlantic ocean water. When Mediterranean sea water enters the Atlantic over the Gibraltar sill, it moves several hundred kilometres into the Atlantic at a depth of about 1000 meters with its own warm, saline, and less dense characteristics. The Mediterranean water stabilises at this depth¹⁹¹⁰ [see Figure 1].

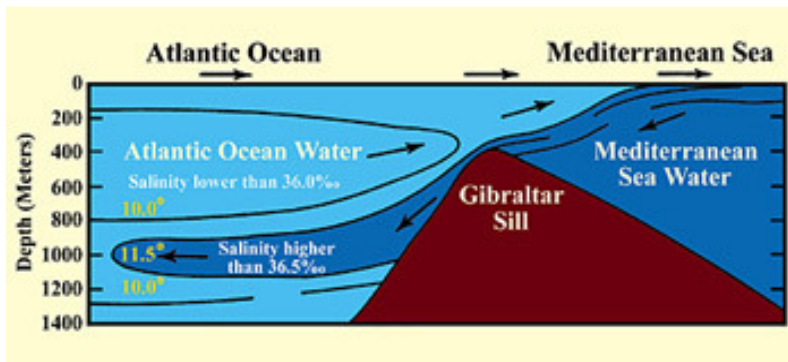


Figure 1: The Mediterranean sea water as it enters the Atlantic over the Gibraltar sill with its own warm, saline, and less dense characteristics, because of the barrier that distinguishes between them. Temperatures are in degrees Celsius (C°). (Marine Geology, Kuenen, p. 43, with a slight enhancement.)

Although there are large waves, strong currents and tides in these seas, they do not mix or transgress this barrier.

The Qur'ān mentioned that there is a barrier between two seas that meet and that they do not transgress. God has said:

¹⁹⁰⁹ Davis, *Principles of Oceanography*, pp. 92-93.

¹⁹¹⁰ Ibid 93.

“He released the two seas, meeting [side by side]; Between them is a barrier [so] neither of them transgresses.”

Qur'ān 55:19-20

But when the Qur'ān speaks about the divider between fresh and salt water, it mentions the existence of “a forbidding partition” with the barrier. God has said in the Qur'ān:

“And it is He who has released [simultaneously] the two seas, one fresh and sweet and one salty and bitter, and He placed between them a barrier and prohibiting partition.”

Qur'ān 25:53

One may ask, why did the Qur'ān mention the partition when speaking about the divider between fresh and salt water, but did not mention it when speaking about the divider between the two seas?

Modern science has discovered that in estuaries, where fresh (sweet) and salt water meet, the situation is somewhat different from what is found in places where two seas meet. It has been discovered that what distinguishes fresh water from salt water in estuaries is a “pycnocline zone with a marked density discontinuity separating the two layers.”¹⁹¹¹ This partition (zone of separation) has a different salinity from the fresh water and from the salt water [see Figure 2].

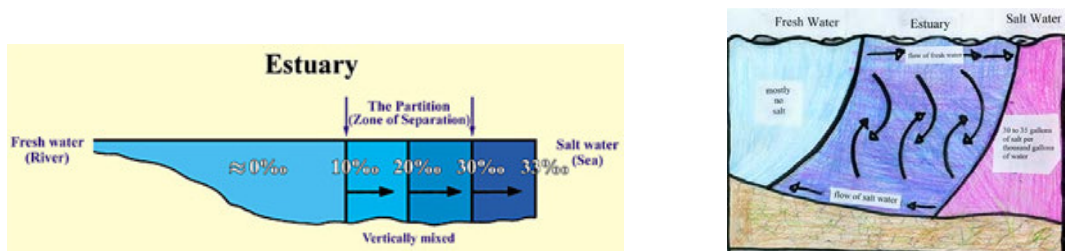


Figure 2: Longitudinal section showing salinity (parts per thousand ‰) in an estuary. We can see here the partition (zone of separation) between the fresh and the salt water.¹⁹¹²

This information has been discovered only recently, using advanced equipment to measure temperature, salinity, density, oxygen dissolubility, etc. The human eye cannot see the difference between the two seas that meet, rather the two seas appear to us as one homogeneous sea. Likewise, the human eye cannot see the division of water in estuaries into the three kinds: fresh water, salt water and the partition (zone of separation).

God has said in the Qur'ān:

“Or [they are] like darkneses within an unfathomable sea which is covered by waves, upon which are waves, over which are clouds - darkneses, some of them upon others. When one puts out his hand [therein], he can hardly see it.”

Qur'ān 24:40

¹⁹¹¹ Gross, *Oceanography*, p. 242.

¹⁹¹² Thurman, *Introductory Oceanography*, pp. 300-301.

This verse mentions the darkness found in deep seas and oceans, where if a man stretches out his hand, he cannot see it. The darkness in deep seas and oceans is found around a depth of 200 meters and below. At this depth, there is almost no light [see Figure 3]. Below a depth of 1000 meters there is no light at all.¹⁹¹³ Human beings are not able to dive more than forty meters without the aid of submarines or special equipment. Human beings cannot survive unaided in the deep dark part of the oceans, such as at a depth of 200 meters.

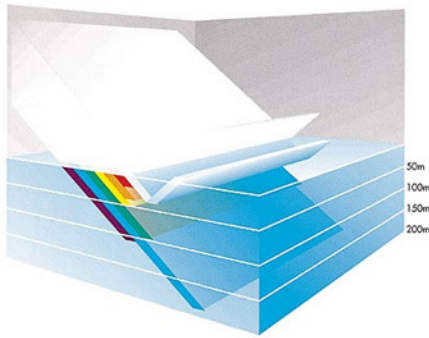


Figure 3: Between 3 and 30 percent of the sunlight is reflected at the sea surface. Then almost all of the seven colours of the light spectrum are absorbed one after another in the first 200 meters, except the blue light. (*Oceans*, Elder and Pernetta, p. 27.)

Scientists have recently discovered this darkness by means of special equipment and submarines that have enabled them to dive into the depths of the oceans.

We can also understand from the following sentences in the previous verse, “...in a deep sea. It is covered by waves, above which are waves, above which are clouds....,” that the deep waters of seas and oceans are covered by waves, and above these waves are other waves. It is clear that the second set of waves are the surface waves that we see, because the verse mentions that above the second waves there are clouds. But what about the first waves? Scientists have recently discovered that there are internal waves which “occur on density interfaces between layers of different densities.”¹⁹¹⁴ [see figure 4].

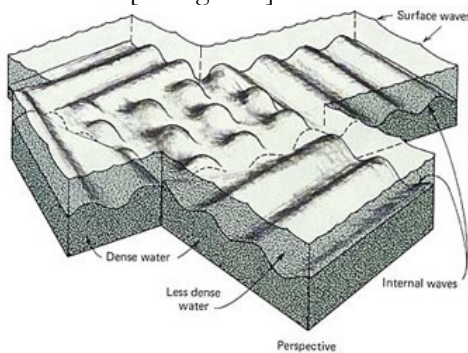


Figure 4: Internal waves at interface between two layers of water of different densities. One is dense (the lower one), the other one is less dense (the upper one).

The internal waves cover the deep waters of seas and oceans because the deep waters have a higher density than the waters above them. Internal waves act like surface waves. They can also break, just like surface waves. Internal waves cannot be seen by the human eye, but they can be detected by studying temperature or salinity changes at a given location.¹⁹¹⁵

¹⁹¹³ *Oceans*, Elder and Pernetta, p. 27.

¹⁹¹⁴ Gross, *Oceanography*, p. 205.

¹⁹¹⁵ *Oceanography*, Gross, p. 205.

Miracle of Iron

Iron is one of the elements highlighted in the Qur'ān. In the chapter known *Al-Hadeed*, meaning Iron, we are informed:

“And We sent down iron, wherein is great military might and benefits for the people, and so that God may make evident those who support Him and His messengers unseen. Indeed, God is Powerful and Exalted in Might.”

Qur'ān 57:25

The word “*anzalna*,” translated as “sent down” and used for iron in the verse, could be thought of having a metaphorical meaning to explain that iron has been given to benefit people. But, when we take into consideration the literal meaning of the word, which is, being physically sent down from the sky, as this word usage had not been employed in the Qur'ān except literally, like the descending of the rain or revelation, we realise that this verse implies a very significant scientific miracle. Because, modern astronomical findings have disclosed that the iron found in our world has come from giant stars in outer space.¹⁹¹⁶

Not only the iron on earth, but also the iron in the entire Solar System, comes from outer space, since the temperature in the Sun is inadequate for the formation of iron. The sun has a surface temperature of 6,000 degrees Celsius and a core temperature of approximately 20 million degrees. Iron can only be produced in much larger stars than the Sun, where the temperature reaches a few hundred million degrees. When the amount of iron exceeds a certain level in a star, the star can no longer accommodate it and it eventually explodes in what is called a ‘nova’ or a ‘supernova.’ These explosions make it possible for iron to be given off into space.

One scientific source provides the following information on this subject: “*There is also evidence for older supernova events: Enhanced levels of iron-60 in deep-sea sediments have been interpreted as indications that a supernova explosion occurred within 90 light-years of the sun about 5 million years ago. 60 is a radioactive isotope of iron, formed in supernova explosions, which decays with a half-life of 1.5 million years. An enhanced presence of this isotope in a geologic layer indicates the recent nucleosynthesis of elements nearby in space and their subsequent transport to the earth (perhaps as part of dust grains).*”¹⁹¹⁷

All this shows that iron did not form on the Earth, but was carried from Supernovas, and was “sent down,” as stated in the verse. It is clear that this fact could not have been known in the 7th century, when the Qur'ān was revealed. Nevertheless, this fact is related in the Qur'ān, the Word of God, Who encompasses all things in His infinite knowledge. The fact that the verse specifically mentions iron is quite astounding, considering that these discoveries were made at the end of the 20th century. In his book *Nature's Destiny*, the well-known microbiologist Michael Denton emphasises the importance of iron:

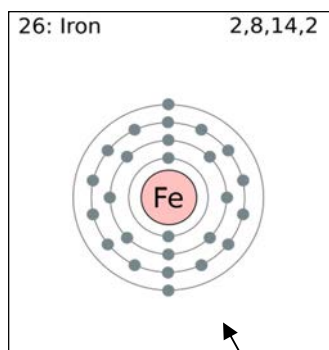
“Of all the metals there is none more essential to life than iron. It is the accumulation of iron in the centre of a star which triggers a supernova explosion and the subsequent scattering of the vital atoms of life throughout the cosmos. It

¹⁹¹⁶ Kazi, Dr. Mazhar U. *130 Evident Miracles in the Qur'an*, p. 110-111;

¹⁹¹⁷ Frisch, Priscilla, *The Galactic Environment of the Sun*, American Scientist, January-February 2000.

was the drawing by gravity of iron atoms to the centre of the primeval earth that generated the heat which caused the initial chemical differentiation of the earth, the out-gassing of the early atmosphere, and ultimately the formation of the hydrosphere. It is molten iron in the centre of the earth which, acting like a gigantic dynamo, generates the earth's magnetic field, which in turn creates the Van Allen radiation belts that shield the earth's surface from destructive high-energy-penetrating cosmic radiation and preserve the crucial ozone layer from cosmic ray destruction...Without the iron atom, there would be no carbon-based life in the cosmos; no supernova, no heating of the primitive earth, no atmosphere or hydrosphere. There would be no protective magnetic field, no Van Allen radiation belts, no ozone layer, no metal to make hemoglobin [in human blood], no metal to tame the reactivity of oxygen, and no oxidative metabolism...The intriguing and intimate relationship between life and iron, between the red colour of blood and the dying of some distant star, not only indicates the relevance of metals to biology but also the bio-centricity of the cosmos..."¹⁹¹⁸

This account clearly indicates the importance of the iron atom. The fact that particular attention is drawn to iron in the Qur'ān also emphasises the importance of the element.



“And We sent down iron, wherein is great military might and benefits for the people, and so that God may make evident those who support Him and His messengers unseen. Indeed, God is Powerful and Exalted in Might.”

Qur'ān 57:25

The Periodic Table of the Elements

1																	2				
H Hydrogen (1.00794)																	He Helium (4.0026)				
3	4															5	6	7	8	9	10
Li Lithium (6.941)	Be Beryllium (9.012182)															B Boron (10.81)	C Carbon (12.011)	N Nitrogen (14.00647)	O Oxygen (15.9994)	F Fluorine (18.9984032)	Ne Neon (20.1797)
11	12															13	14	15	16	17	18
Na Sodium (22.98976928)	Mg Magnesium (24.304)															Al Aluminum (26.9815385)	Si Silicon (28.0855)	P Phosphorus (30.973761)	S Sulfur (32.066)	Cl Chlorine (35.4527)	Ar Argon (39.948)
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
K Potassium (39.0983)	Ca Calcium (40.078)	Sc Scandium (44.955912)	Ti Titanium (47.867)	V Vanadium (50.9415)	Cr Chromium (51.9961)	Mn Manganese (54.938044)	Fe Iron (55.845)	Co Cobalt (58.9332)	Ni Nickel (58.6934)	Cu Copper (63.546)	Zn Zinc (65.39)	Ga Gallium (69.723)	Ge Germanium (72.6300)	As Arsenic (74.9216)	Se Selenium (78.96)	Br Bromine (79.904)	Kr Krypton (83.80)	Rb Rubidium (85.4678)	Sr Strontium (87.62)	Y Yttrium (88.90585)	Zr Zirconium (91.224)
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58
Rb Rubidium (85.4678)	Sr Strontium (87.62)	Y Yttrium (88.90585)	Zr Zirconium (91.224)	Nb Niobium (92.90638)	Mo Molybdenum (95.94)	Tc Technetium (98.9062)	Ru Ruthenium (101.07)	Rh Rhodium (102.9055)	Pd Palladium (106.42)	Ag Silver (107.8682)	Cd Cadmium (112.411)	In Indium (114.818)	Sn Tin (118.710)	Sb Antimony (121.757)	Te Tellurium (127.60)	I Iodine (126.90549)	Xe Xenon (131.29)	Ba Barium (137.327)	La Lanthanum (138.90547)	Ce Cerium (140.12)	Pr Praseodymium (140.90768)
55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76
Cs Cesium (132.90545)	Ba Barium (137.327)	La Lanthanum (138.90547)	Hf Hafnium (178.49)	Ta Tantalum (180.94788)	W Tungsten (183.84)	Re Rhenium (186.207)	Os Osmium (190.23)	Ir Iridium (192.222)	Pt Platinum (195.084)	Au Gold (196.96657)	Hg Mercury (200.59)	Tl Thallium (204.3833)	Pb Lead (207.2)	Bi Bismuth (208.9804)	Po Polonium (209)	At Astatine (210)	Rn Radon (222)	Fr Francium (223)	Ra Radium (226)	Ac Actinium (227)	Th Thorium (232)
87	88	89	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122
Fr Francium (223)	Ra Radium (226)	Ac Actinium (227)	Rf Rutherfordium (261)	Db Dubnium (262)	Sg Seaborgium (263)	Bh Bohrium (264)	Hs Hassium (265)	Mt Meitnerium (266)	Ds Darmstadtium (267)	Rg Roentgenium (268)	Cn Copernicium (269)	Nh Nihonium (270)	Fl Flerovium (271)	Mc Moscovium (272)	Lv Livermorium (273)	Ts Tennessine (274)	Og Oganesson (276)	La 			

58

Ce

Thorium
Th
(90.0239)

59

Pr

Praseodymium
Pr
(90.90748)

60

Nd

Neodymium
Nd
(144.24)

61

Pm

(145)

62

Sm

Samarium
Sm
(150.36)

63

Eu

Euroium
Eu
(151.964)

64

Gd

Gadolinium
Gd
(157.25)

65

Tb

Terbium
Tb
(158.92535)

66

Dy

Dysprosium
Dy
(162.50015)

67

Ho

Holmium
Ho
(164.93033)

68

Er

Erbium
Er
(167.25594)

69

Tm

Thulium
Tm
(168.93403)

70

Yb

Ytterbium
Yb
(173.04538)

71

Lu

Lutetium
Lu
(174.9674)

90

Th

Thorium
Th
(90.0239)

91

Pa

Protactinium
Pa
(91.02608)

92

U

Uranium
U
(92.0233)

93

Np

Neptunium
Np
(92.90638)

94

Pu

Plutonium
Pu
(93.94215)

95

Am

Americium
Am
(94.06394)

96

Cm

Curium
Cm
(95.0432)

97

Bk

Berkelium
Bk
(96.06069)

98

Cf

Californium
Cf
(97.04637)

99

Es

Einsteinium
Es
(98.07232)

100

Fm

Fermium
Fm
(99.0711)

101

Md

Mendelevium
Md
(100.1281)

102

No

Nobelium
No
(101.0778)

103

Lr

Lr

Zaghloul El-Naggar, professor of earth science and geology, has an interesting theory. He concludes that the Qur'ānic account of the celestial origin of iron is coupled by another miraculous aspect which is represented by the fact that the number of both the Qur'ānic chapter on iron and of the

¹⁹¹⁸ Denton, Michael J., *Nature's Destiny*, p.198.

verse that mentions this element in the same chapter precisely correspond to both the atomic weight (55.847 or roughly 56) and the atomic number (26) of iron, respectively.

Indeed the number of Sūrah al-Hadeed (the Qur'ānic Chapter on Iron) is '57' and the number of the verse is '25,' but the Qur'ān in its text separates its introduction (Surat Al-Fātihah or the Opening) from the rest of the Book, and considers the 'Basmalah' (In the Name of God, the Most Compassionate, the Most Merciful) as a Qur'ānic verse at the beginning of this Sūrah (Al-Fātihah) and of every other Qur'ānic Sūrah where it is mentioned.

Taking this Qur'ānic direction into consideration, the number of Sūrah al-Hadeed becomes '56' which is the closest figure to the atomic weight of the most abundant iron isotope (55.847), and the number of the verse becomes '26' which is the **exact atomic number** of iron. The existence of an iron isotope with the atomic weight of 57 also fits very well with the current numbering of the chapter on iron in the Glorious Qur'ān. [God knows better]

The Qur'ān on Mountains

The book entitled 'Earth' is a basic reference textbook in many universities around the world. One of its two authors is Professor Emeritus Frank Press. He was the Science Advisor to former US President Jimmy Carter, and for 12 years was the President of the National Academy of Sciences, Washington, DC. His book says that mountains have underlying roots.¹⁹¹⁹ These roots are deeply embedded in the ground, thus, mountains have a shape like a peg [see Figures 1, 2 and 3].

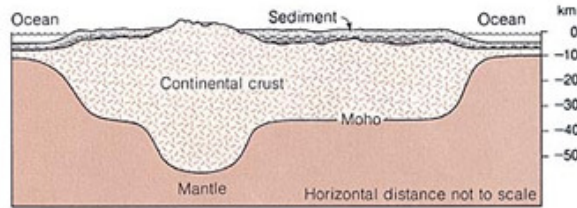


Figure 1: Mountains have deep roots under the surface of the ground.¹⁹²⁰

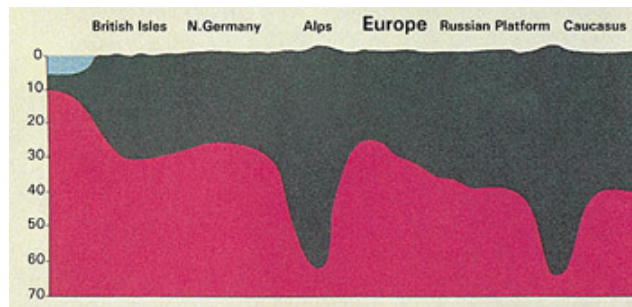


Figure 2: Schematic section. The mountains, like pegs, have deep roots embedded in the ground.¹⁹²¹

¹⁹¹⁹ Press, E.F., *Earth*, p. 435.

¹⁹²⁰ Ibid, p. 413.

¹⁹²¹ Cailleux, *Anatomy of the Earth*, p. 220.

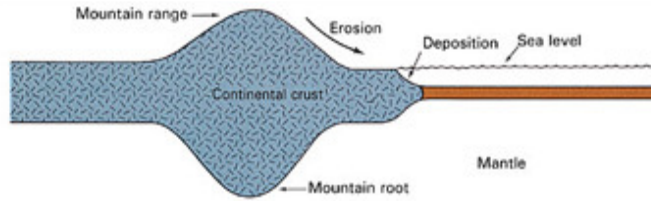


Figure 3: Another illustration shows how the mountains are peg-like in shape, due to their deep roots.¹⁹²²

This is how the Qur'ān has described mountains. God has said in the Qur'ān:

“Have We not made the earth a resting place? And the mountains as stakes (pegs)?”

Qur'ān 78: 6-7

Modern earth sciences have proven that mountains have deep roots under the surface of the ground [see Figure 3] and that these roots can reach several times their elevations above the surface of the ground.¹⁹²³ So the most suitable word to describe mountains on the basis of this information is the word ‘peg,’ since most of a properly set peg is hidden under the surface of the ground. The history of science tells us that the theory of mountains having deep roots was introduced only in the latter half of the nineteenth century.¹⁹²⁴

Mountains also play an important role in stabilising the crust of the earth.¹⁹²⁵ They hinder the shaking of the earth. God has said in the Qur'ān:

“And He has cast into the earth firmly set mountains, lest it shift with you, and [made] rivers and roads, that you may be guided.”

Qur'ān 16: 15

Likewise, the modern theory of plate tectonics holds that mountains work as stabilisers for the earth. This knowledge about the role of mountains as stabilisers for the earth has just begun to be understood in the framework of plate tectonics since the late 1960's.¹⁹²⁶

Could anyone during the time of the Prophet Muhammad have known of the true shape of mountains? Could anyone imagine that the solid massive mountain which he sees before him actually extends deep into the earth and has a root, as scientists assert? A large number of books of geology, when discussing mountains, only describe that part which is above the surface of the earth. This is because these books were not written by specialists in geology. However, modern geology has confirmed the truth of the Qur'ānic verses.

¹⁹²² Tarbuck and Lutgens, *Earth Science*, p. 158.

¹⁹²³ Naggār, El-, *The Geological Concept of Mountains in the Qur'ān*, p. 5.

¹⁹²⁴ Ibid, p.5.

¹⁹²⁵ Ibid p.44 -45.

¹⁹²⁶ Ibid p.5.

The Qur'ān on Human Embryonic Development

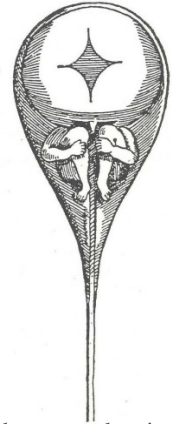
Historic Preamble

Western physicians, during the 16th century, believed that a human being's origin began entirely in the male's semen. They envisaged that inside the male semen was a miniscule, fully formed human being who grew larger once implanted into the womb of the mother.

During the 17th and 18th centuries, the previously held view was replaced by the opinion that the 'minuscule' human instead formed out of the woman's menstrual blood coagulating [thickening] and that the male semen was merely an activating agent. It was not until 1775, that Italian scientist Spallanxi established that both the male semen and female ovum were required to form the human embryo.

It was also not until modern times, with the help of the electron microscope, that scientists discovered the existence of chromosomes and genes and that human embryonic development takes place in successive distinct stages.

The terminology used to describe human development in the Qur'ān is characterized by descriptiveness and accuracy. Until recently these statements were not fully appreciated, since they referred to details in human development which were scientifically unknown in earlier times.



17th century drawing of a sperm by Hartsoecker

In the Qur'ān, God mentions that human embryonic development passes through a number of distinct stages:

“...We placed him as a sperm-drop [*nutfah*] in a firm lodging [i.e., womb]. Then We made the sperm-drop into a clinging clot [*‘alaqah*], and We made the clot [*‘alaqah*] into a [*mudghah*] lump [of flesh], and We made [from] the lump, [*idham*] bones, and We covered the bones with [*lahm*] flesh; then We developed him into another creation. So blessed is God, the best of creators.”

Qur'ān 23:12-14

The stages in the Ayat [verse] can be summarised as follows:

Stage 1: The Nutfah

Stage 2: The ‘Alaqah

Stage 3: The Mudghah

Stage 4: Bone Formation [Idham]

Stage 5: Clothing the Bones with Flesh [Lahm]

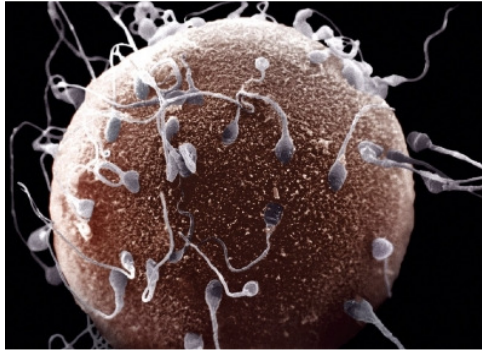
Stage 1: The Nutfah

The *Nutfah* literally means ‘a [single] drop’ of fluid whereas *Manii* means ‘semen.’ God says, **“Had he not been a sperm [*nutfah*] from semen [*manii*] emitted?”** [Qur’ān 75:36]

In the Qur’ān and Hadith, Nutfah is used in three different but related contexts:

1. The Male Nutfah [Qur’ān 75:36]
2. The Female Nutfah
3. Nutfah Amshaj – mixed or mingled Male and Female Nutfah [Qur’ān 76:2]

The Male Nutfah



The word Nutfah was mentioned twelve different times in the Qur’ān and the word Manii was mentioned thrice. In the male context, the Nutfah is a **single particle** from the Manii when it is ejaculated – i.e. a single cell [sperm] from amongst the 200-300 million sperm cells. Before the 16th century, writings on embryonic development did not distinguish the constituent parts of the semen in its role of fertilisation.

“Does man think that he will be left neglected? Had he not been a sperm [*nutfah*] from semen [*manii*] emitted? Then he was a clinging clot [*‘alaqah*], and [God] created [his form] and proportioned [him]. And made of him two mates, the male and the female. Is not that [Creator] Able to give life to the dead?”

Qur’ān 75:36-40

“And that He creates the two mates - the male and female - From a sperm-drop [*nutfah*] when it is emitted.”

Qur’ān 53:45-46

In these Ayaat [verses], there are a lot of facts that require careful consideration. We know the sex of the newborn is determined by the sperm - it is definitively stated that the male and female are fashioned from a sperm-drop from the semen that has been ejaculated. If a sperm carrying an X chromosome fertilises an ovum [which always contains an X chromosome], the offspring will be a girl, while if the fertilising sperm contains a Y chromosome, the offspring will be a boy.

The Qur’ān has stated this fact 1,400 years ago, before anybody knew anything about X and Y chromosomes.

The Female Nutfah

The Female Nutfah [ovum] per se is not mentioned explicitly in the Qur'ān, but is inferred in the term Nutfah Amshaj – i.e. mingled from both male and female [refer to Qur'ān 76:2]. However, it is clearly stated in the hadith from the Prophet.

A Jew came to the Prophet and asked, 'O Muhammad. Tell me from what thing man is created.' The Prophet said, 'O Jew. From both Male and Female Nutfah, man is created.'¹⁹²⁷

This is a very astonishing revelation, as it is only recently that we came to know that both male and female cells [sperm and ovum] join together to form the human zygote – a fact not known before the 19th century.

The Role of Genes

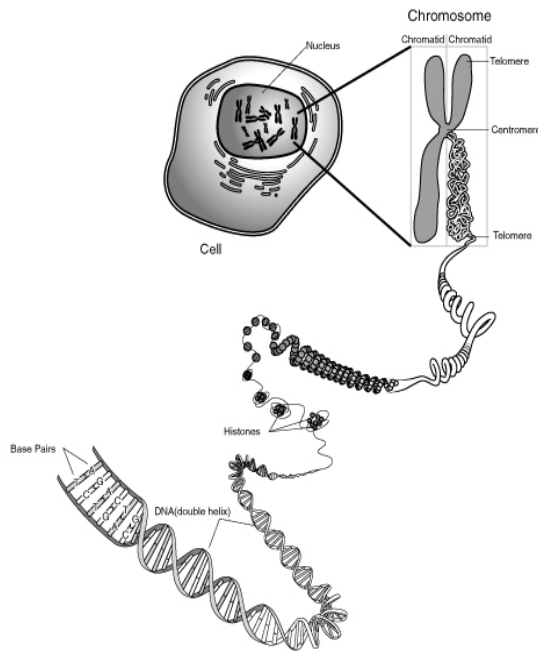
'From what substance did He create him? From a sperm-drop [*nutfah*] He created him and destined [*qadr*] for him.'

Qur'ān 80:18-19

In a single male Nutfah [sperm cell], which measures 60 microns [1 micron = 1/1000 mm], there are 23 chromosomes – long spiral double strand helices. They contain the genes which determine every type of characteristic the body has, like hair colour, skin type, etc.

Hence, it is within this Nutfah that God has determined and destined all the physical characteristics that an individual has. It is amazing that the description of these realities are so accurately stated by God in the Qur'ān.

The Prophet said, 'God has ordained an angel that accompanies the different stages of development of the Nutfah. The 'Alaqa, the Mudgha and in every stage he asks God, 'O God, what to do next?' If God determines its full development, the angel asks, 'Is it a boy or a girl? Happy or unhappy, his livelihood and his life span. All is written [determined] while he is in the mother's womb.' [Bukhari]



¹⁹²⁷ Musnad Ahmad. The complete narration has been declared *da'eef* [contains weakness according to the rules of hadeeth verification] by Ahmad Shakir.

Stage 2: The 'Alaqah Stage

Linguistic Analysis

According to many Arabic dictionaries, the word '*alaqah*' includes the following meanings:

1. Attached and hanging to something,
2. Blood clot,
3. Leech.

1. 'Alaqah as 'attached and hanging'

As we see in Figure 1, the embryo [which is represented by the bilaminar embryonic disc] is attached to the placenta and is hanging or suspended in the chorionic cavity by the connecting stalk. This is in agreement with the meaning of the word '*alaqah*' as "attached and hanging to something."

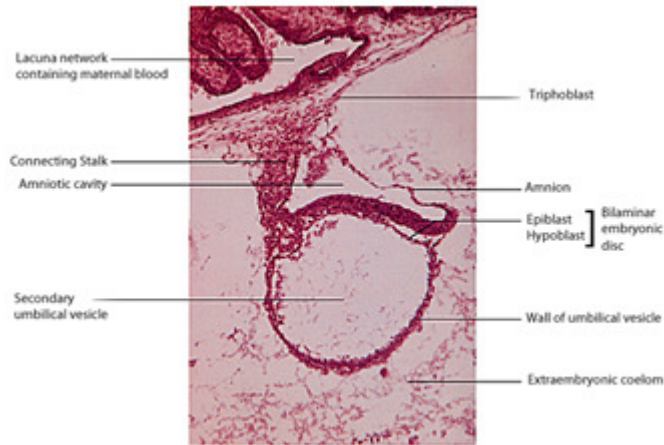


Figure 1: Photomicrographs of longitudinal sections of an implanted embryo at approximately 14 days. High-power view [Å—95]. The embryo is represented by the bilaminar embryonic disc composed of epiblast and hypoblast. [From Nishimura H [ed]: *Atlas of Human Prenatal Histology*. Tokyo, Igaku-Shoin, 1983]. The embryo is now attached to the primitive placenta and is suspended or hanging via the 'connecting stalk.'¹⁹²⁸

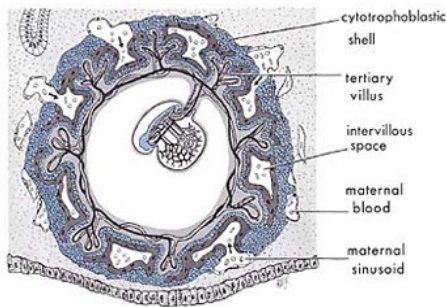


Figure 2: We can see in this diagram the suspension of an embryo during the '*alaqah*' stage in the womb [uterus] of the mother. Moore and Persaud, *The Developing Human*, 5th ed., p. 66.

¹⁹²⁸ Kareem, E., *Embryology in the Qur'an: The Alaqah Stage*, www.islampapers.com.

2. 'Alaqah as 'blood clot'

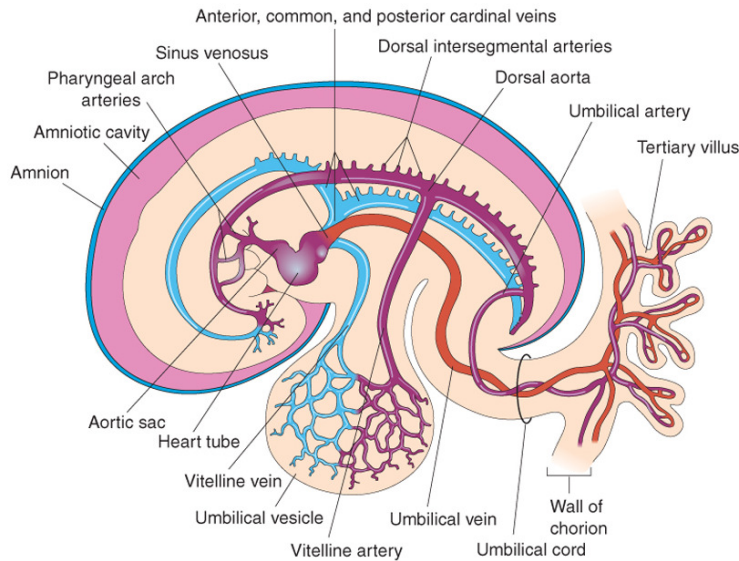


Figure 3: Diagram of the primordial cardiovascular system in an embryo of approximately 21 days, viewed from the left side during the 'alaqah stage. The external appearance of the embryo and its sacs is similar to that of a blood clot, due to the presence of relatively large amounts of blood in the embryo and the chorion. The umbilical vein carries well-oxygenated blood and nutrients from the chorion sac to the embryo. The umbilical arteries carry poorly oxygenated blood and waste products from the embryo to the chorion. [From Moore and Persaud [2007].

Figure 3 shows a diagram of the primitive cardiovascular system in an embryo of about 21 days. During this stage we find that the external appearance of the embryo and its sacs is similar to that of a blood clot.

“Implantation begins at about the 6th to 7th day after fertilization. The part of the blastocyst projecting into the uterine cavity remains relatively thin. The syntrophoblast contains a proteolytic enzyme which causes destruction of the endometrial cells so that that the blastocyst sinks deeper and deeper into the uterine mucosa...The final deficiency in the endometrium is sealed off by **a blood or fibrin clot**, overlying the blastocyst. This cover is called the operculum. By about 10 to 12 days after fertilization, the blastocyst is completely encased in the endometrium and thus, implantation is complete.”¹⁹²⁹

The blood, though fluid, does not circulate until the end of the third week. On the 21st day, the heart of the embryo connects with the blood vessels in the embryo, the connecting stalk, the chorion and the umbilical vesicle [yolk sac], and the blood starts to circulate and the heart begins to beat. Thus, the embryo takes the appearance of a blood clot even though its blood is fluid.

¹⁹²⁹ Allan, J., & Kramer, B. *The Fundamentals of Human Embryology*, [2nd ed.], p. 23: Wits University Press – as quoted by Kareem, E., *Embryology in the Qur'ān: The Alaqah Stage*.

3. 'Alaqah as 'leech'

Scholars, linguists and dictionaries have all mentioned one of the meanings of 'alaqah as a leech¹⁹³⁰. The fourteenth century dictionary *Lisān al-'Arab* states that “'alaqah refers to a worm living in the water that sucks blood, the plural of which is 'alaq”¹⁹³¹ and in the dictionary of *al-Qāmūs al-Muhit* that 'alaq is “a small creature of water that sucks blood [a leech].”¹⁹³² The word 'alaqah also occurs in several languages related to Arabic. In Hebrew there is עֲלֻקָּה 'alūqāh [or alukah]¹⁹³³, the generic name for any blood-sucking worm or leech. And in Aramaic and Syriac there are words with apparently similar meanings. In Ad-Damīrī's Arabic zoological lexicon, *Hayāt al-Hayawān* [The Life of the Animals, 1372 C.E.], there is an article on the leech ['alaq]¹⁹³⁴ and in Ibn Wahshīya's *Kitāb al-Sumūm* [The Book on Poisons, c. 950 C.E.] there is the treatment for the one who has swallowed a leech ['alaq].¹⁹³⁵

A popular ninth century Christian polemic against Islam claims that Muslims believe that “God created man from a leech” based on the work of Nicetas of Byzantium. Nicetas, who wrote between 842 and 867 C.E., had a copy of the Qur'ān in Greek translation which he made use of to identify the tenets of Islam. His Greek translation renders both 'alaq and 'alaqah as *bdella* [βδέλλα], meaning “leech.”¹⁹³⁶

The classic Qur'ānic commentator, Ibn Kathīr [b. 1302 C.E.], mentions the meaning of “elongated like the shape of a leech - العلقة مستطيلة”¹⁹³⁷. Finally, *The Qur'ān: an Encyclopedia* has an entry for 'alaq that also mentions the same meanings: “The linguistic definition of 'alaq [singular 'alaqa] is 'leech', 'medicinal leech', '[coagulated] blood', 'blood clot', or 'the early stage of the embryo.’”¹⁹³⁸

¹⁹³⁰ This section is an edited version of Kareem, E., *Embryology in the Qur'an: The Alaqah Stage*, www.islampapers.com.

¹⁹³¹ Ibn Manẓūr, in *Lisān al-'Arab*, Dār Ṣādir, Beirut, n.d., vol. 10, pp 261-268; as cited in Zindani et al. [1994, p. 68].

¹⁹³² *Al-Qāmūs al-Muhit*, vol. 3, p 275 as cited in Zindani et al. [1994, p. 68].

¹⁹³³ “The leech [עֲלֻקָּה 'alūqāh] has two daughters: Give and Give.” Proverbs 30:15 [ESV]. Hebrew עֲלֻקָּה 'alūqāh meaning a leech. [Blue Letter Bible] Although the Hebrew word is translated leech in most versions of the Bible, there has been much dispute whether this is the proper meaning. Recourse is therefore to the Arabic language - See Kaltner, The Use of Arabic in Biblical Hebrew Lexicography: Catholic Biblical Association of America. [1996, pp. 86-87].

¹⁹³⁴ *Kitāb Hayāt al-hayawān* [The Book of the Lives of the Animals] finished in 1372 C.E. as mentioned in De Somogyi [1950, p. 42].

¹⁹³⁵ Ibn Wahshīya's Book on Poisons c.950 C.E. Known under various titles: *Kitāb al-Shānāq fī al-Sumūm wa'al-tiryāq*, *Kitāb al-Sumūm wa'al-tiryāqāt*, and *al-Sumūm wadaḡ' madarrbā*. Levey [1966, p. 84].

¹⁹³⁶ “Nicetas accuses the Qur'ān of teaching that man comes from a leech [Confutatio1, lines 90–92]: λέγει, ὅτι ἐκ βδέλλης ὁ ἄνθρωπος γίνεται [he says that man is created from a leech]. The phrase is then picked up by Zigabenos, who finds it absurd...” Simelidis [2011, pp. 900-902] *The Byzantine Understanding of the Qur'ānic Term al-Ṣamad and the Greek Translation of the Qur'ān. Speculum*, 86(04), 887-913.

¹⁹³⁷ Ibn Kathīr, *Tafsīr Al-Qur'ān Al-'Aẓīm*, p. 242 (1st ed. Vol. 3). 1980. Beirut, Lebanon: Dār Al-Fikr.

¹⁹³⁸ Sahin, H. “Alaq” p. 27. In O. Leaman [Ed.] [2006], *The Qur'ān: An Encyclopedia*. Routledge.

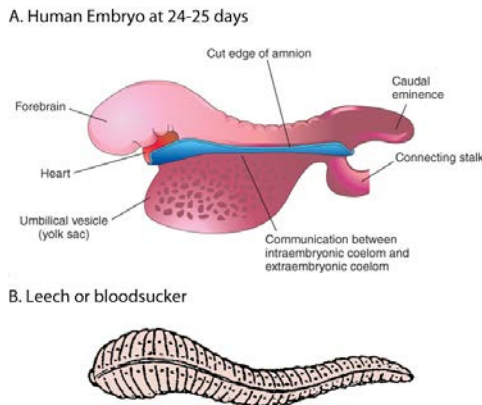


Figure 4. Drawings illustrating the similarities in appearance between a human embryo and a leech [*‘alaqah*]. **A**, shows a lateral view of an embryo [size 2.5-3.0mm] at days 24 to 25 during folding, showing the large forebrain and the ventral position of the heart [from Moore & Persaud: *The Developing Human* 9th Edition [2013]. **B**, shows a drawing of a leech.

Note the leech-like appearance of the human embryo at this stage.

A leech is an apt description of the early human embryo. The embryo clings to the endometrium or lining of the uterus [day 7] just as a leech clings to the skin. The embryo is also surrounded by amniotic fluid just as the leech is surrounded by water. If we consider the literal meaning of “leech” for *‘alaqah*, we find that during the third week, the embryo loses its round shape and elongates until it takes the shape of a leech.

Figure 4 above and Figure 5 clearly indicate that the shape of the embryo does in fact resemble a leech. At this stage the cardiovascular system has started to appear and the embryo is now dependent upon the maternal blood for its nutrition like a leech which feeds on the blood of others.¹⁹³⁹



Figure 5 **A**, shows a lateral view of an embryo [size 2.5-3.0mm] at days 24 to 25 [Modified from Moore & Persaud: *The Developing Human* 8th Edition]. **B**, *Hirudo medicinalis*, medicinal leech **C**, Scanning electron micrograph of an embryo at Week 4, 26 - 30 days [Professor Kathy Sulik, The University of North Carolina]. Note the leech-like appearance of the human embryos at this stage.

¹⁹³⁹ The umbilical vein carries well-oxygenated blood and nutrients from the chorion sac to the embryo. The arteries carry poorly oxygenated blood and waste products to the chorionic villi for transfer to the mother's blood.

In the BBC television series, *The Human Body: The Incredible Journey from Birth to Death*, Professor Robert Winston¹⁹⁴⁰ also describes the embryo in a similar way. Prof. Winston demonstrates how the embryo obtains nourishment from the blood of the mother by comparing it with a leech which feeds on the blood of others,

“[The leech] takes whatever it needs to live by sucking the blood of whatever it can latch onto; in this case that’s me! As it sucks my blood, it takes from it all that it needs to live, it literally lives off me and the whole of pregnancy is shaped by a similar kind of parasitic relationship...it does raid her blood for the raw materials it needs to grow. From the word go, both leech and embryo are out for themselves.”¹⁹⁴¹



Figure 6 Presenter Professor Robert Winston with a blood sucking leech [*‘alaqah*] attached to his forearm. Professor Winston shows how the embryo obtains nourishment from the blood of the mother, similar to the leech which feeds on the blood of others. [*The Human Body. The Incredible Journey from Birth to Death*, © BBC Worldwide Ltd, 1998].

Similarly, in *Anatomy Demystified*, the early embryo is described as worm-like in appearance which is nourished by the mother’s maternal blood supply, “Another membrane becomes the *yolk sac*, which provides nourishment for the early embryo. By 24 days, a *connecting stalk* appears in the middle of the now **worm-like body**.”¹⁹⁴²

A segmented body like a leech

The body of the leech is divided into a number of segments which gives rise to a ringed appearance of the body, hence the name “ringed worms.”¹⁹⁴³ The human embryo is also segmented just like a leech or worm as Professor Peter Nathanielsz describes in *A Time to be Born: The Life of the Unborn Child*, “By the end of the third week the **embryo has undergone segmentation, rather like an earth worm**, and now consists of zones like stacked circular tires.”¹⁹⁴⁴

¹⁹⁴⁰ Prof. Robert Winston is Professor of Science and Society and Emeritus Professor of Fertility Studies at Imperial College, London.

¹⁹⁴¹ Prof Winston in *The Human Body: The Incredible Journey from Birth to Death*, BBC Worldwide [1998].

¹⁹⁴² Layman, D. P., *Anatomy Demystified*, p. 366, [2004] New York: McGraw-Hill Professional; London: McGraw-Hill.

¹⁹⁴³ Garwood, P. R., & Campbell, A. [2007]. “Segmented Worms.” The Encyclopedia of Underwater Life. Oxford Reference Online.

¹⁹⁴⁴ Nathanielsz [1994, p. 22]. Peter W. Nathanielsz is a Professor at the Laboratory for Pregnancy and Newborn Research, Cornell University, Ithaca, USA. “Professor Nathanielsz was amongst the handful of

These layers curl to form a tube-like structure which Anthony Smith, in *The Human Body*, also likens to a worm, “the early embryo is like a worm, with a gut running from one end to the other, an outer covering also running from end to end and a central layer filling the space between the two.”¹⁹⁴⁵ Ted Zerucha in *Human Development* also describes the gut of the embryo as a tube, “Running through the body, along the anterior-posterior axis, is the gut. The gut is essentially a tube that runs from the mouth, through the digestive system, to the anus.”¹⁹⁴⁶ The tube-like depiction of the embryo’s gut is not unlike that of an annelid as described in *The Columbia Encyclopedia*, “The digestive system of annelids consists of an unsegmented gut that runs through the middle of the body from the mouth, located on the underside of the head, to the anus, which is on the pygidium [the posterior terminal region].”¹⁹⁴⁷

Internal structure of a leech

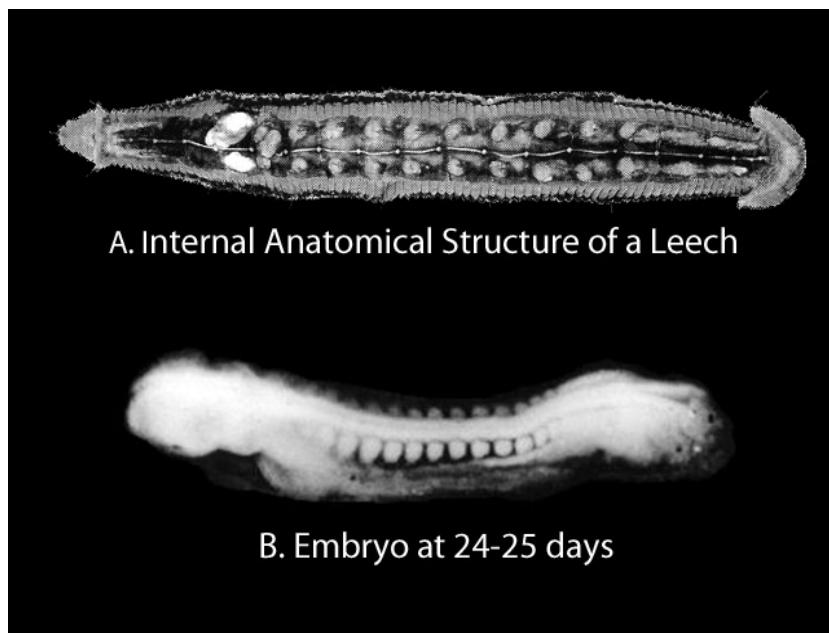


Figure 7 A, Ventral dissection showing the internal anatomical structure of a leech. [From J.G. Nicholls and D. Van Essen. *The nervous system of the leech*, 1974, Scientific American 230:38-48]. **B**, Dorsal view of a 13-somite embryo at approximately 24 days, actual size 3.0mm. [From Professor Hideo Nishimura, Kyoto University, Kyoto, Japan]. Note the remarkable similarity in appearance between the human embryo and the internal structure of the leech.

pioneers who assisted at the birth about thirty years ago of the new discipline of fetology and has remained at the forefront of what is now an enormous field. His laboratory has contributed many of the technical advances that now allow the most intimate details of fetal life to be examined with a precision equal to that of a cosmologists’ radio-telescope.”

¹⁹⁴⁵ Smith, *The Human Body: The Incredible Journey from Birth to Death*, p. 38.

¹⁹⁴⁶ Zerucha, T., *Human Development*, p. 52.

¹⁹⁴⁷ “Annelida” in *The Columbia Encyclopedia* [2008].

If we examine the anatomy of the leech we find that the appearance of its internal structures is also similar to that of the human embryo:

Figure 7 A shows a ventral view of a dissected medicinal leech. Note how the body is made up of a number of similar segments which resemble the somites in human embryos.

Figure 7 B shows an embryo at 24-25 days. The actual size of the embryo at this stage is just 3.0mm.

Figure 8 A-C showing dorsal views of embryos during the third and fourth weeks.

Figure 8 D Shows of the internal structure of the leech. Note the remarkable similarity in appearance between the embryos and the anatomy of the leech.

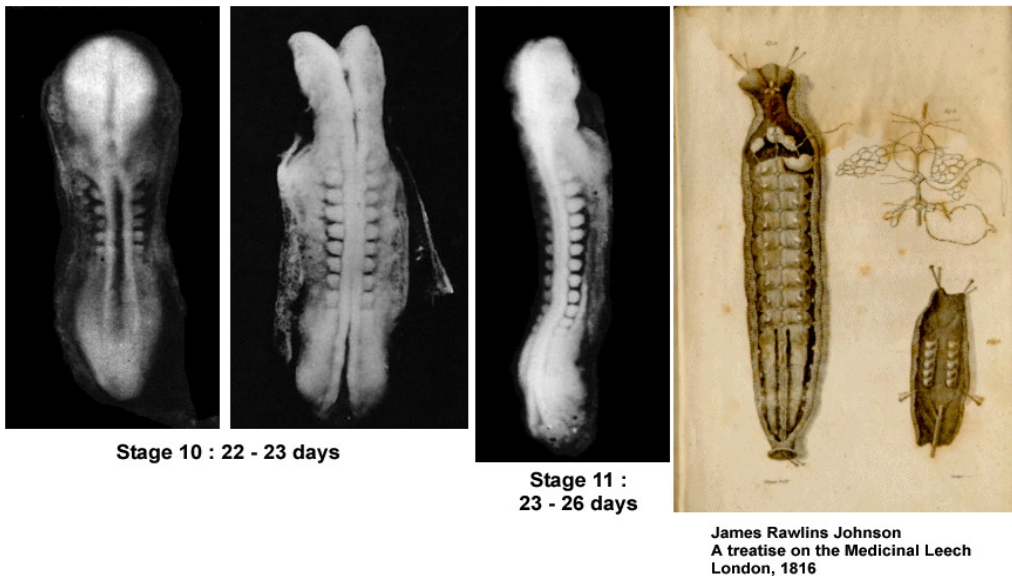


Figure 8 Dorsal views of embryos during the third and fourth weeks. **A**, Dorsal view of a 5-somite embryo, actual size 2.5mm. **B**, Dorsal view of an older eight-somite embryo, actual size 3.0mm. **C**, Dorsal view of a 13-somite embryo at approximately 24 days, actual size 3.0mm [Photographs from Professor Hideo Nishimura, Kyoto University, Kyoto, Japan]. **D**, The anatomical structure of the leech [Illustrated by James Rawlins Johnson, *A Treatise on the Medicinal Leech*, London, 1816. [Rare – In process] UCLA Biomedical Library: History and Special Collections for the Sciences].

Prevention of blood-clotting

A striking similarity between the leech and the embryo is the way in which enzymes are released to facilitate easy blood-flow and the prevention of the clotting of blood. As the embryo draws nourishment [leech-like] from the mother's blood, the anticoagulant enzyme, Thrombomodulin, [TM] prevents the blood clotting. In the leech, the protein that serves the same function is called Hirudin.

Summary of 'Alaqah stage

The Qur'ānic term *'alaqah* is a comprehensive expression for the second stage of embryonic development that descriptively encompasses the primary external and internal features. In this one word, the general shape of the embryo as a leech is described, the internal events such as the formation of blood and closed vessels are described, and the attachment of the embryo to the placenta is also brought to mind.

The similarity between the embryo and leech is remarkable:

- the external shape of the leech resembles an embryo at 22-25 days [Figure 4 and 5],
- the internal structure of the leech resembles an embryo of 22-26 days [Figure 7 and 8],
- the embryo clings to the lining of the uterus in a similar way to a leech that clings to the skin,
- the embryo obtains nourishment from the blood of the mother [Figure 3], like the leech which feeds on the blood of others [Figure 6],
- the embryo has a segmented body like a worm or leech,
- the early embryo further resembles a leech in that it has a tube-like gut running from one end to the other.

The Qur'ānic term *'alaqah* refers to the embryo when it is extremely small. The *'alaqah* is just 0.7-3.0mm in length. Due to the small sizes involved, scientists could not have recognised the detailed features of the *'alaqah* stage until the second half of the 19th century and the beginning of the 20th.

Prof. Keith L. Moore concludes that it is, “remarkable how much the embryo of 23-24 days resembles a leech. As there were no microscopes or lenses available in the 7th century, doctors would not have known that the human embryo had this leech-like appearance. In the early part of the fourth week, the embryo is just visible to the unaided eye because it is smaller than a kernel of wheat.”¹⁹⁴⁸

¹⁹⁴⁸ Moore, K. L., *A Scientist's Interpretation of References to Embryology in the Qur'ān*.

Stage 3: Mudghah stage



Figure 9. Photograph of embryo at the end of the *'alaqah* stage [age 24 to 25 days]. Ten pairs of the 13 pairs of somites are easily recognized, but the embryo is still relatively straight and has a leech-like appearance.

The embryo at 24-25 days is finishing the *'alaqah* stage. It changes into the *mudghah* stage at 26-27 days. The transformation from *'alaqah* to *mudghah* is in fact very rapid, and during the last day or two of the *'alaqah* stage, the embryo is beginning to develop some of the characteristics of the *mudghah*, e.g. the somites begin to appear and become a distinct feature of this stage.

One of the meanings of the word *mudghah* is “something that is chewed by teeth.” If one were to take a piece of gum and chew it in his or her mouth and then compare it with an embryo at the *mudghah* stage, we would conclude that the embryo at the *mudghah* stage acquires the appearance of a chewed substance. This is because of the somites at the back of the embryo that “somewhat resemble teeth-marks in a chewed substance.”¹⁹⁴⁹ [see Figures 10 and 11].

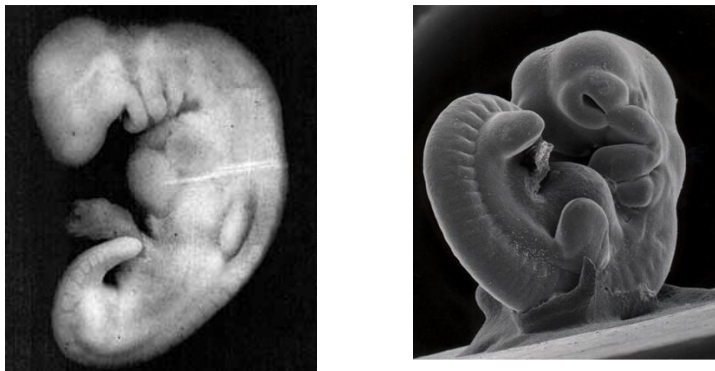


Figure 10: Top left photograph of an embryo at the *mudghah* stage [28 days old]. The embryo at this stage acquires the appearance of a chewed substance, because the somites at the back of the embryo somewhat resemble teeth marks in a chewed substance. The actual size of the embryo is 4 mm.¹⁹⁵⁰

¹⁹⁴⁹ Moore and Persaud, *The Developing Human*, 5th ed., p. 8.

¹⁹⁵⁰ Moore and Persaud, *The Developing Human*, 5th ed., p. 82 – from Professor Hideo Nishimura, Kyoto University, Japan.

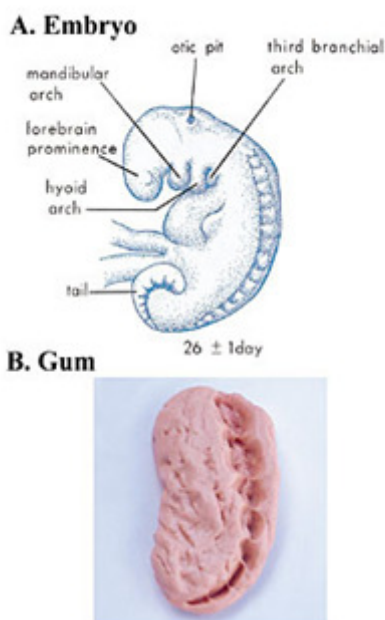


Figure 11: When comparing the appearance of an embryo at the mudghah stage with a piece of gum that has been chewed, we find similarity between the two.

A. Drawing of an embryo at the mudghah stage. We can see here the somites at the back of the embryo that look like teeth marks. [Moore and Persaud, *The Developing Human*, 5th ed., p. 79.]

B. Photograph of a piece of gum that has been chewed.

The appearance of the somites or “imprints” changes continuously, just as the teeth imprint changes on a chewed substance with each act of chewing. The embryo changes its overall shape, but the structures derived from the somites remain. Just as a substance acquires furrows, swellings and a corrugated surface as it is being chewed, so does the appearance of the embryo.

The embryo turns in its position due to the modifications in its centre of gravity with new tissue formation, similar to the turning of a substance with chewing.

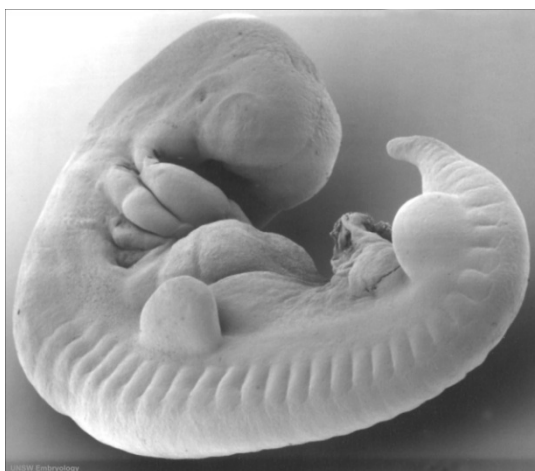


Figure 12: Human embryo day 32 with 35 somites. The embryo is about 7.0 mm in length. Note the indentations that are identified between somites, and with these indentations, the embryo resembles a chewed substance in its external appearance.

In Figure 12 above, the embryo looks somewhat like a chewed lump. The chewed appearance results from the somites which resemble teeth marks. The somites [cuboidal blocks of mesodermal tissue] represent the beginnings or primordia of the vertebrae. By the 3rd week of human embryonic development, about 38 pairs somites form. By the 5th week there are 42-44 pairs of somites. Most of the axial skeleton [skull, vertebral column, ribs, and sternum] and skeletal muscles will be derived from these somites.¹⁹⁵¹

As there were no microscopes available in the 7th century C.E., people would not have known that the human embryo had this chewed-like appearance. Professor Marshall Johnson states:

“You have to be really careful on what is the definition of ‘seeing.’ I can see a piece of dandruff on this table top; I can just barely make it out because this is a nice black surface [but] I can see no detail in it. If I want to see detail in it then I need some sort of visual aid, something to aid my vision, I need a magnifying glass, I need a microscope. So I might be able to see a piece of dandruff, but to see any detail in it as is described in the Qur’ān, I need an instrument that wasn’t developed until the 1700s.”¹⁹⁵²

Stage 4: Bone Formation [Idham]

God continues, **“...and We made [from] the lump [mudghah], bones, and We covered the bones with flesh”** [23:14]. The mudghah or somite embryo is fashioned into bones which are clothed with flesh.

Hamilton, Boyd and Mossman write that “the somites are the bases from which the greater part of the axial skeleton and musculature develop.”¹⁹⁵³

The timing of this phase has been mentioned in the following statement of the Prophet Muhammad, “When 42 nights [i.e. 6 weeks] have passed from the time of the nutfah [time of conception], God sends an angel to it, who shapes it and makes its ears, eyes, skin, muscles and bones...”¹⁹⁵⁴

“Before the 42nd day, it is difficult to distinguish the human embryo from the embryos of many animals, but at this time it becomes clearly distinguishable in its appearance.”¹⁹⁵⁵ The formation of the skeleton gives the embryo its human shape.

¹⁹⁵¹ This section is an edited version of Kareem, E., *Embryology in the Qur'an: The Mudghah Stage and Some observations on the mudghah (chewed-like) stage of human development*, www.islampapers.com.

¹⁹⁵² As quoted in Kareem, E., *Some observations on the mudghah stage of human development*, www.islampapers.com.

¹⁹⁵³ Hamilton, Boyd, Mossman, *Human Embryology*, 4th Edition, as quoted in Albar, M.Ali, *Human Development as revealed in the Holy Qur'an and Hadith*, p. 79.

¹⁹⁵⁴ Sahih Muslim, *Kitab al-Qadr*.

¹⁹⁵⁵ G.C. Goeringer, A. A. Zindani, M. A. Ahmed, *Embryology in the Qur'an: Bone and Muscle Development*, www.islampapers.com.

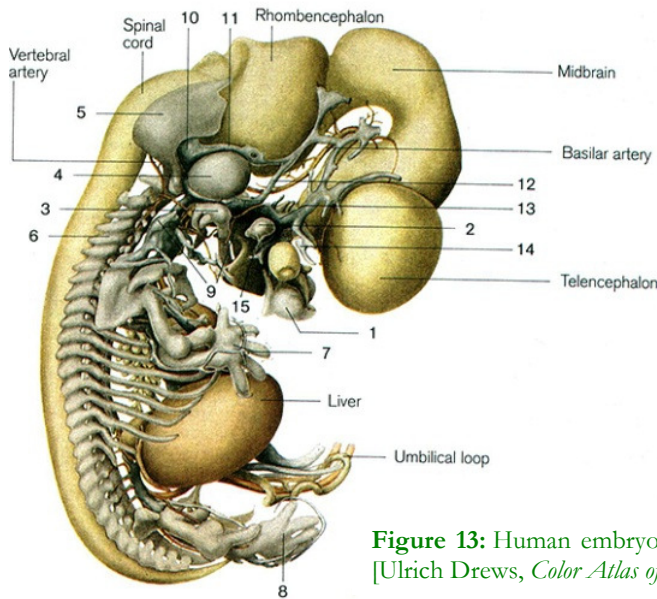


Figure 13: Human embryo at about days 50-51 days
[Ulrich Drews, *Color Atlas of Embryology*, 1995]

In the 6th week the cartilaginous skeleton begins to form and the embryo acquires a soft skeleton [as we see in Figure 13]:

“Formation of bone does not begin uniformly throughout the body. Rather, there is a sequential appearance of bony tissue. However, in the 7th week the spreading development of the skeleton occurs. Bone development in the limbs commences in the limb buds from mesenchymal cells. Primary ossification centres appear in the femur during week 7 and in the sternum [breast bone] and the maxilla [upper jaw] in weeks 8-9.”¹⁹⁵⁶

Stage 5: Clothing the Bones with Flesh [Lahm]

God says, “**...We covered the bones with [lahm] flesh**” [23:14]. In the *Fundamentals of Human Embryology*, it is noted that, “Soon after the cartilaginous models of the bones have been established, the myogenic cells, which have now become myoblasts, aggregate to **form muscle masses** on the ventral [front] and dorsal [back or posterior] aspects of the limbs.”¹⁹⁵⁷ Although precursor cells [myoblasts, or primitive muscle cells] are present adjacent to developing bone, “differentiation into skeletal muscle attachments occur **after** the ossification process in the shaft and ends of the bones has begun.”¹⁹⁵⁸

¹⁹⁵⁶ Ibid.

¹⁹⁵⁷ Allan, J., and Kramer, B., *The Fundamentals of Human Embryology*, 2nd Edition, Wits University Press, 2010, p. 148.

¹⁹⁵⁸ G.C. Goeringer, A. A. Zindani, M. A. Ahmed, *Embryology in the Qur'an: Bone and Muscle Development*, www.islampapers.com.

Conclusion

It is clear from the preceding pages that God, in the Qur'ān, gives a detailed account of the development of the human embryo. Firstly, it accurately describes the main stages of development. Each word describes the characteristic of a specific stage and its morphological and physiological identity. Secondly, it describes the sequence of these events in the same chronological order as discovered by the electron microscope.

Scientists Acceptance of the Miracles of the Qur'ān

Professor Emeritus Keith L. Moore is one of the world's most prominent scientists in the fields of anatomy and embryology and is the author of the book entitled 'The Developing Human', which has been translated into eight languages. Dr. Keith Moore is Professor Emeritus of Anatomy and Cell Biology at the University of Toronto, Toronto, Canada. There, he was Associate Dean of Basic Sciences at the Faculty of Medicine and for 8 years was the Chairman of the Department of Anatomy. In 2007, Professor Moore became the first recipient of the Henry Gray/Elsevier Distinguished Educator Award, The American Association of Anatomists' (AAA) highest award for human anatomy education. Most recently in 2012, Prof. Moore received the Queen's Diamond Jubilee Medal – a commemorative medal to honour significant contributions and achievements by Canadians.

Professor Moore said,

“....Because the staging of human embryos is complex, owing to the continuous process of change during development, it is proposed that a new system of classification could be developed using the terms mentioned in the Qur'ān and Sunnah. The proposed system is simple, comprehensive, and conforms with present embryological knowledge. The intensive studies of the Qur'ān and hadeeth [reliably transmitted reports of the Prophet Muhammad] in the last four years have revealed a system for classifying human embryos that is amazing since it was recorded in the 7th Century C.E. Although Aristotle, the founder of the science of embryology, realised that chick embryos developed in stages from his studies of hen's eggs in the fourth century B.C., he did not give any details about these stages. As far as it is known from the history of embryology, little was known about the staging and classification of human embryos until the twentieth century. For this reason, the descriptions of the human embryo in the Qur'ān cannot be based on scientific knowledge in the seventh century. The only reasonable conclusion is: these descriptions were revealed to Muhammad from God. He could not have known such details because he was an illiterate man with absolutely no scientific training.”¹⁹⁵⁹

Consequently, Professor Moore was asked the following question: “Does this mean that you believe that the Qur'ān is the word of God?” He replied: “I find no difficulty in accepting this.”¹⁹⁶⁰

Professor Moore consulted a number of embryologists for their opinions:

¹⁹⁵⁹ *This is the Truth* - www.islam-guide.com/truth.htm

¹⁹⁶⁰ Ibid

“...I was invited to Saudi Arabia to lecture on embryology at King AbdulAzziz University in Jeddah, Saudi Arabia... and while I was there, at my suggestion, invited Dr. [T.V.N] Persaud and Dr. [E. Marshall] Johnson to come to Saudi Arabia. And they [the Embryology Committee] asked them the same questions, and I purposefully didn't tell them my interpretations, I wanted them to give their own. So they did and their answers were similar to mine...Dr. Johnson is one of the most outstanding embryologist and teratologist in the United States. We didn't just pick anyone, I picked the best. And in Canada, Dr. Persaud at the University of Manitoba, where I spent twenty years, is also an outstanding embryologist who has three doctors degrees... so I picked the very best. And then, when I started thinking about other embryologists around the world, we brought in Dr. [Robert] Edwards from Cambridge [world-renowned for his early work on in vitro fertilization]... so we invited him to Saudi Arabia and again he was asked the same questions and they [the Embryology Committee] got essentially the same answers.... [and] one of my colleagues in Kyoto, Japan....he didn't go to Saudi Arabia, [and] he has been consulted and so on. So we have consulted embryologists around the world for their opinions on these statements in the Qur'an, and it's clear from what Dr. Persaud has said and from all of our work in this area that these statements [in the Qur'an] are correct.”¹⁹⁶¹

Professor Emeritus T. V. N. Persaud is Professor of Anatomy, Professor of Pediatrics and Child Health and Professor of Obstetrics, Gynecology and Reproductive Sciences at the University of Manitoba, Winnipeg, Manitoba, Canada. There, he was the Chairman of the Department of Anatomy for 16 years. He is well-known in his field. He is the author or editor of 22 textbooks and has published over 181 scientific papers. In 1991, he received the most distinguished award presented in the field of anatomy in Canada, the J.C.B. Grant Award from the Canadian Association of Anatomists. Henry Gray/Elsevier Distinguished Educator Award, American Association of Anatomists, 2010.

When he was asked about the scientific miracles in the Qur'an, he stated the following:

“The way it was explained to me is that Muhammad was a very ordinary man. He could not read, didn't know [how] to write. In fact, he was an illiterate. And we're talking about twelve [actually about fourteen] hundred years ago. You have someone illiterate making profound pronouncements and statements and that are amazingly accurate about scientific nature. And I personally can't see how this could be a mere chance. There are too many accuracies and like Dr. Moore, I have no difficulty in my mind that this is a divine inspiration or revelation which led him to these statements.”¹⁹⁶²

Professor Persaud has included some Qur'anic verses and sayings of the Prophet Muhammad in some of his books.

Dr. E. Marshall Johnson is Professor Emeritus of Anatomy and Developmental Biology at Thomas Jefferson University, Philadelphia, Pennsylvania, USA. There, for 22 years he was Professor of Anatomy, the Chairman of the Department of Anatomy, and the Director of the

¹⁹⁶¹ Prof. Moore lecture to Muslim Students Association, “Embryology in the Qur'an with Drs. Persaud, Moore and Johnson [1988]”, <http://youtu.be/ZJRRhfK5xUI?t=37m32s>

¹⁹⁶² Taken from the video documentary, *This is Truth.*

Daniel Baugh Institute. He was also the President of the Teratology Society. He has authored more than 200 publications. In 1981, during the Seventh Medical Conference in Dammam, Saudi Arabia, Professor Johnson said in the presentation of his research paper:

“Summary: The Qur'ān describes not only the development of external form, but emphasises also the internal stages, the stages inside the embryo, of its creation and development, emphasising major events recognised by contemporary science.”

Also he said: “As a scientist, I can only deal with things which I can specifically see. I can understand embryology and developmental biology. I can understand the words that are translated to me from the Qur'ān. As I gave the example before, if I were to transpose myself into that era, knowing what I knew today and describing things, I could not describe the things which were described.

I see no evidence for the fact to refute the concept that this individual, Muhammad, had to be developing this information from some place. So I see nothing here in conflict with the concept that divine intervention was involved in what he was able to write.”¹⁹⁶³

Dr. Yoshihide Kozai is Professor Emeritus at Tokyo University, Hongo, Tokyo, Japan, and was the Director of the National Astronomical Observatory, Mitaka, Tokyo, Japan. He said:

“I am very much impressed by finding true astronomical facts in [the] Qur'ān, and for us the modern astronomers have been studying very small pieces of the universe. We've concentrated our efforts for understanding of [a] very small part.

Because by using telescopes, we can see only very few parts [of] the sky without thinking [about the] whole universe. So, by reading [the] Qur'ān and by answering the questions, I think I can find my future way for investigation of the universe.”¹⁹⁶⁴

Professor Tejatat Tejasen is the Chairman of the Department of Anatomy at Chiang Mai University, Chiang Mai, Thailand. Previously, he was the Dean of the Faculty of Medicine at the same university. During the Eighth Saudi Medical Conference in Riyadh, Saudi Arabia, Professor Tejasen stood up and said:

“During the last three years, I became interested in the Qur'ān From my study and what I have learned from this conference, I believe that everything that has been recorded in the Qur'ān fourteen hundred years ago must be the truth, that can be proved by the scientific means.

Since the Prophet Muhammad could neither read nor write, Muhammad must be a messenger who relayed this truth, which was revealed to him as an enlightenment by the one who is eligible [as the] creator. This creator must be God.

¹⁹⁶³ Taken from the video documentary, *This is Truth.*

¹⁹⁶⁴ Ibid.

Therefore, I think this is the time to say *La ilaha illa Allāh*, there is no deity to worship except God, *Muhammadur rasoolu Allāh*, Muhammad is Messenger [Prophet] of Allāh [God]. Lastly, I must congratulate for the excellent and highly successful arrangement for this conference . . . I have gained not only from the scientific point of view and religious point of view but also the great chance of meeting many well-known scientists and making many new friends among the participants. The most precious thing of all that I have gained by coming to this place is *La ilaha illa Allāh*, *Muhammadur rasoolu Allāh*, and to have become a Muslim.”¹⁹⁶⁵

After all these examples we have seen about the scientific miracles in the Holy Qur'ān, let us ask ourselves these questions:

Could it be just a coincidence that all this recently discovered scientific information from different fields was mentioned in the Qur'ān, which was revealed fourteen centuries ago? Could this Qur'ān have been authored by Muhammad or by any other human being? How could Prophet Muhammad have possibly known all this 1,400 years ago, when scientists have only recently discovered this using advanced equipment and powerful microscopes which did not exist at that time?

The only answer is that the Qur'ān is the word of God Almighty.

¹⁹⁶⁵ Taken from the video documentary, *This is Truth?*